2618. 02, 03, 05, 57, 70, 71, 74, 0, 89, 90, 91, 98, 106, 107, 121, 125,

CHAPTER – 8

D.C. MOTORS

8.1 Introduction:

A direct current motor converts electrical energy into mechanical energy. It is similar in construction to a D.C. generator. Hence, a D.C. motor can be used as a D.C. generator and vice-versa. The external appearance of a D.C. motor may be slightly different from that of the D.C. generator, mainly due to the fact that, the frame of the generator may be partially open, as it is usually located in a clean environment and operated only by skilled workers. A motor is always operated in a dusty environment and usually unskilled workers operate them. Hence, the frames of the motors are usually closed.

8.2 Working Principle:

A D.C. motor works on the principle that *"whenever a current carrying conductor is placed in a magnetic field, it experiences a force"*. The magnitude of the force experienced by the conductor is given by,

$$F = B I \ell, \qquad (8.1)$$

Where, F = Force experienced in Newtons

B = Flux density of the magnetic field in Wb/m^2

I = Current flowing through the conductor in amperes

ℓ = Length of the conductor in metres

The direction of the force is given by Fleming's left hand rule.

8.3 Fleming's Left Hand Rule:

It states that *"when the thumb, fore finger and the middle finger of the left hand are held mutually perpendicular to each other, the fore finger in the direction of the magnetic field, the middle finger in the direction of the current, then the direction of the thumb indicates the direction of the force experienced by the conductor"*.

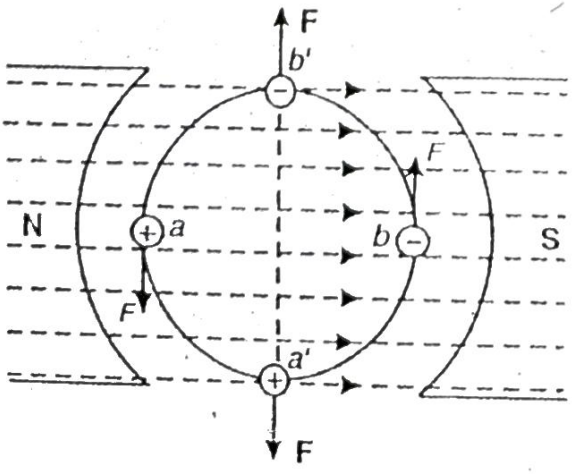


Fig.8.1

Consider a D.C. motor having two poles north and south represented by N and S as shown in Fig. 8.1. There will be conductors placed uniformly in the slots of the armature.

For the sake of explaining the a principle of working of a D.C. motor, only two conductors a and b, which are placed under the influence of N-pole and S-pole respectively and which are connected together by an end connection at the back side of the armature and to the

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For a cumulatively compounded D.C. motor, as the load increases I_a also increases and the resultant flux is given by,

$$\phi = \phi_{sh} + \phi_{se}$$

As I_a increases, ϕ_{se} increases and hence, the resultant flux also increases. Hence, due to increase of I_a, the speed decreases as I_aR_a increases and it is further decreased due to increase of ϕ_{se}. Hence, the speed decreases faster than a D.C. shunt motor and slower than a D.C. series motor. The variation of N with respect to I_a is as shown in Fig.8.16.

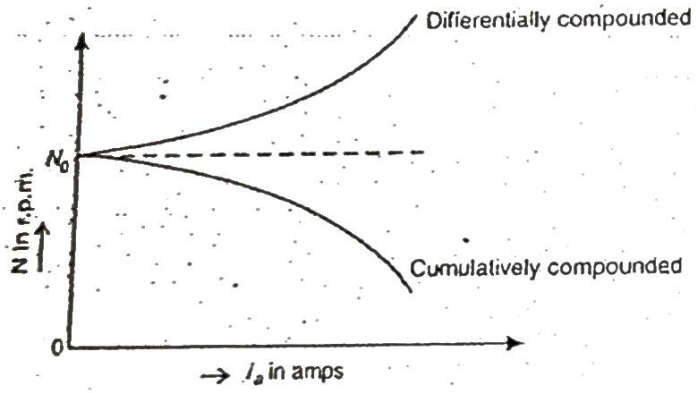


Fig.8.16

In the case of a differentially compounded motor, as I_a increases, the speed decreases due to increase of I_aR_a, but the resultant flux $\phi = \phi_{sh} - \phi_{se}$ decreases due to which the speed increases. The decrease of speed due to increase of I_aR_a is very small compared to the increase of speed due to the decrease of ϕ. Hence, with the increase of I_a, the speed actually increases. This characteristic of the speed increasing on load is also an undesirable characteristic.

(c) N / T_a Characteristic:

We know that, for a D.C. motor, $T_a \propto \phi I_a$ (8.35)

For a cumulatively compounded motor as the load increases, I_a also increases. As I_a increases, the flux $\phi = \phi_{sh} + \phi_{se}$ also increases. Hence, T_a increases faster and the speed also decreases faster.

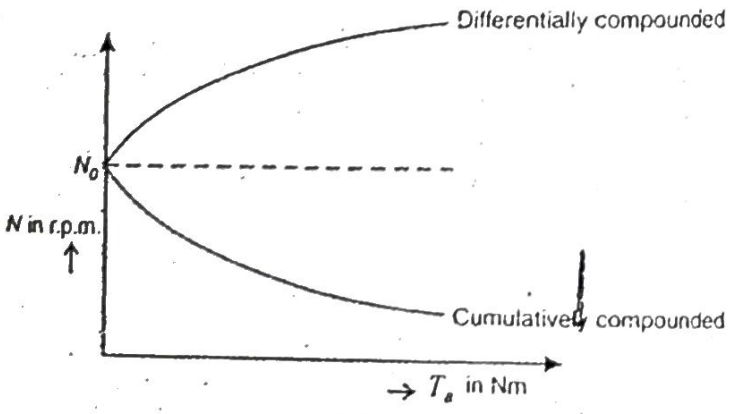


Fig.8.17

For a differentially compounded motor, as the load increases, I_a also increases. As I_a increases, the flux $\phi = \phi_{sh} - \phi_{se}$ decreases and hence torque T_a decreases faster and the speed also increases faster. The plot of N with respect to T_a is as shown in Fig. 8.17.

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25| 67, 69. 70, 71, 77, 92, 97,
101, 106, 117, 120, 122, 125, 126

commutator segments at front side of the armature are considered. When a D.C. supply is given to the motor terminals, the current flows through the conductors a and b via the commutator. In conductor a, the +ve sign marked indicates that the current is flowing inwards and the –ve sign in conductor b indicates that the current is flowing outwards. The direction of the magnetic field is represented by the lines of magnetic force, which emanate from the north pole N and go into the south pole S as shown in the Fig. 8.1.

According to Fleming's left hand rule, the conductor a experiences a force F in the downward direction and the conductor b, experiences an equal force F in the upward direction. As the two conductors are connected together, the two equal and opposite forces F acting on them, constitute a couple, tending to rotate the armature in anti-clockwise direction. Due to the action of this couple, let the armature rotate by 90° in the anti-clockwise direction and the conductors a and b occupy positions a' and b' respectively.

In this position, they experience a force F in opposite directions along the same line and hence the torque experienced by them is zero. This position is known as "dead center" for the conductors a and b. If the armature contains only these two conductors, then the armature would stop in the position a'b'. But the armature consists of several other conductors which are uniformly distributed in the slots of the armature, which are connected together and experiencing a torque in the anti-clockwise direction. Thus the armature continues to rotate in the anti-clockwise direction.

For the armature to experience a continuous anti-clockwise torque, it is necessary that the directions of currents in conductors a and b must be reversed, after they cross the positions a' and b' respectively. Otherwise, the armature experiences a torque in the clockwise direction, thereby producing a pulsating torque in the dead centre position a' b'. This reversal of current in conductors a and b after they cross positions a' and b' respectively, is effected by the commutator and thus makes the armature to experience a continuous anti-clockwise torque and makes it rotate continuously in the anti-clockwise direction.

7|9 61, 65, 70, 75, 82, 83, 84
90, 93, 98, 102, 103, 111, 22, 25,

8.4 Back E.M.F. (E_b):

Fig. 8.2 symbolically represents a D.C. shunt motor. V is the applied voltage, due to which a current I_a flows through the armature conductors. I_L is the line current and I_{sh} is the current flowing through the shunt field winding

$$I_L = I_a + I_{sh} \qquad \text{-(8.2)}$$

When a current I_a flows through the armature conductors, a torque is produced and the armature rotates. The current I_{sh} flowing through the shunt field winding produces a flux ϕ and hence, an E.M.F. E_b is induced in the armature conductors. The direction of this induced E.M.F. is such as to oppose the applied voltage. Hence, this induced E.M.F. E_b is

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called as the *back E.M.F.* The directions of the applied voltage V and the back E.M.F. E_b are shown in fig.8.2

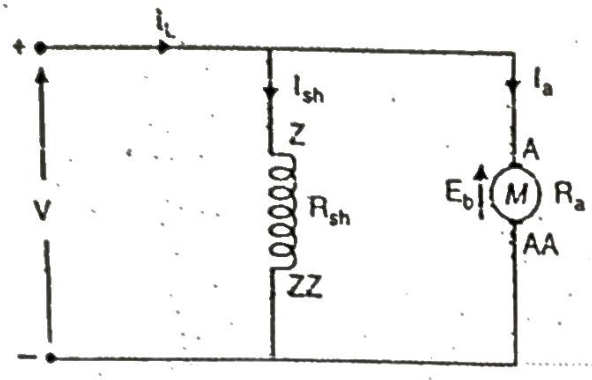


Figure.8.2

The applied voltage V has to drive current through the armature conductors against the opposition of the back E.M.F. and hence work has to be done. This work done is manifested in the form of mechanical power developed by the armature. In the absence of back E.M.F., no mechanical power can be developed by the armature.

The armature current I_a is given by,

$$I_a = \frac{V - E_b}{R_a} \qquad (8.3) \qquad \text{or} \qquad V = E_b + I_a R_a \qquad (8.4)$$

The back E.M.F. is nothing but the induced E.M.F. and hence its equation is,

$$E_b = \frac{\phi ZNP}{60A} \qquad (8.5)$$

Multiplying equation (8.4) by I_a, we get, $\quad V I_a = E_b I_a + I_a^2 R_a \qquad (8.6)$

Where, $V I_a$ = Electrical power input to the armature.

$I_a^2 R_a$ = Copper loss in the armature.

and $E_b I_a$ = Electrical equivalent of the mechanical power developed by the armature, which includes iron losses and mechanical losses.

The efficiency of the D.C. motor is given by

$$\eta = \frac{\text{Mechanical power developed by the armature}}{\text{Electrical power input to the motor}} = \frac{E_b I_a}{V I_a} = \frac{E_b}{V} \qquad (8.7)$$

Higher the value of E_b, higher will be the motor efficiency. The mechanical power developed by the armature is given by,

$$P_m = VI_a - I_a^2 R_a, \qquad \text{i.e.} \qquad \frac{dP_m}{dI_a} = V - 2 I_a R_a$$

For maximum power to be developed, $\frac{dP_m}{dI_a} = 0$

$$\therefore \; V - 2 I_a R_a = 0 \qquad \text{or} \qquad I_a R_a = \frac{V}{2} \qquad (8.8)$$

But, $E_b + I_a R_a = V$

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$$\therefore E_b + \frac{V}{2} = V \quad \text{or} \quad E_b = \frac{V}{2} \tag{8.9}$$

But in practice, D.C. motors are not designed to satisfy the conditions given in equations (8.8) and (8.9), because, if $E_b = V / 2$, the current flowing through the armature will be far more than the rated current and lot of input power will be wasted in the-form of heat, thus increasing the losses and the efficiency of the D.C. motor for the condition $E_b = V/2$ will be less than 50%.

8.5 Torque Equation:

Torque is the turning moment about an axis. It is equal to the product of the force and the radius at which it acts.

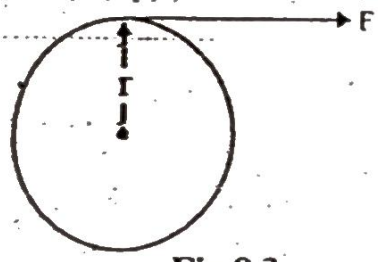


Fig.8.3

Consider the armature of the D.C. motor to have a radius r and let F be the force acting tangential to its surface as shown in Fig. 8.3. The torque exerted by the force F on the armature is given by,

$$T_a = F \times r \ \text{Nm} \tag{8.10}$$

The work done by this force F, in one revolution is given by,

$$W = \text{Force} \times \text{distance covered in one revolution} = F \times 2\pi r \ \text{W–S} \tag{8.11}$$

The power developed by the armature = Work done in one second

$$= F \times 2\pi r \times \text{number of revolutions per second}$$

$$= F \times 2\pi r \times \frac{N}{60} = \frac{2\pi N}{60}(F \times r) = \frac{2\pi N T_a}{60} \ \text{watts} \tag{8.12}$$

We have already learnt in section 8.4 of this chapter that, the electrical equivalent of the mechanical power developed by the armature of the D.C. motor is equal to $E_b I_a$.

$$\therefore \frac{2\pi N T_a}{60} = E_b I_a = \frac{\phi Z N P}{60A} I_a$$

$$\therefore T_a = \frac{1}{2\pi} \phi Z I_a \left(\frac{P}{A}\right) \text{Nm} = 0.159 \, \phi Z I_a \left(\frac{P}{A}\right) \text{Nm} \tag{8.13}$$

$$= 0.0163 \, \phi Z I_a \left(\frac{P}{A}\right) \text{Kgm} \tag{8.14}$$

Equation (8.13) gives the *gross torque* developed by the armature, which includes *iron losses* and *mechanical losses* of the motor. The actual torque available at the shaft to do useful work, which is known as *shaft torque* or *useful torque* T_{sh}, is less than T_a by an amount of torque which is equivalent to iron losses and mechanical losses in the D.C. motor.

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$$\therefore \quad T_{sh} = T_a - T_L \qquad (8.15)$$

Where, T_{sh} = Shaft torque

T_a = Armature torque

T_L = Torque lost due to iron losses and mechanical losses

The useful torque or shaft torque is given by,

Output of the motor in watts $= \dfrac{2\pi N T_{sh}}{60}$ as per equation (8.12)

$$\therefore \quad T_{sh} = \frac{\text{Output of the motor in watts}}{2\pi\frac{N}{60}} \text{ Nm} \qquad (8.16)$$

The output of a D.C motor is usually expressed in H.P.

$$\therefore \quad T_{sh} = \frac{\text{Output in H.P.} \times 735.5}{2\pi\frac{N}{60}} \text{ Nm} \qquad (8.17)$$

8.6 Types of D.C. motors:

Depending on the way in which the field windings are connected to the armature, D.C. motors are classified into three types.

(1) D.C. shunt motor (2) D.C. series motor and (3) D.C. compound motor

The D.C. compound motor may be further classified into two types:

(a) *Cumulatively* compounded D.C. motor, which may be connected either as *long shunt* or *short shunt.*

(b) *Differentially* compounded D.C. motor, which may be connected either as *long shunt* or *short shunt.*

8.7 D.C. Shunt Motor:

Fig.8.4 represents a D.C. shunt motor. In this type of motor, the shunt field winding is connected across the armature. V is the applied voltage due to which a current I_L flows through the line, a current I_{sh} through the shunt field winding and I_a through the armature conductors. The shunt field current is given by,

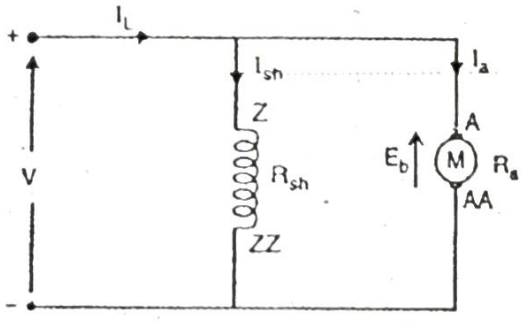


Fig.8.4

$$I_{sh} = \frac{V}{R_{sh}} \qquad (8.18)$$

The armature current is given by,

$$I_a = I_L - I_{sh} \qquad (8.19)$$

The back E.M.F. induced in the armature is given by,

$$E_b = V - I_a R_a - \text{B.C.D.} - \text{A.R.D.} \qquad (8.20)$$

Where, B.C.D. = Brush contact drop and A.R.D. = Armature reaction drop.

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8.8 D.C. Series Motor:

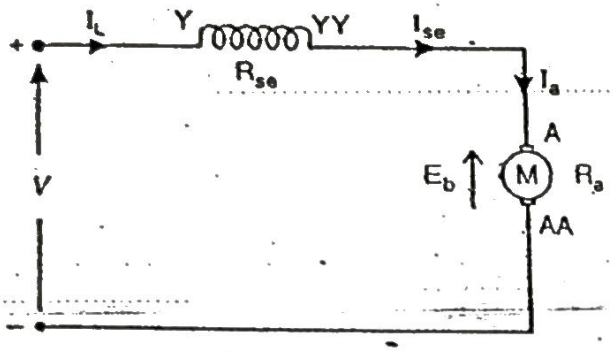


Fig. 8.5

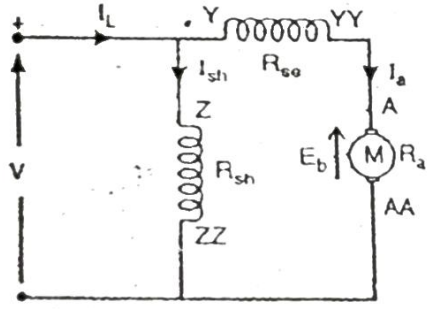
Fig. 8.5 represents a D.C. series motor. In this type of motor, the series field winding is connected in series with the armature. V is the applied voltage due to which a current I_L flows through the line, the series field winding and also through the armature conductors.

$$\therefore \quad I_L = I_{se} = I_a \qquad (8.21)$$

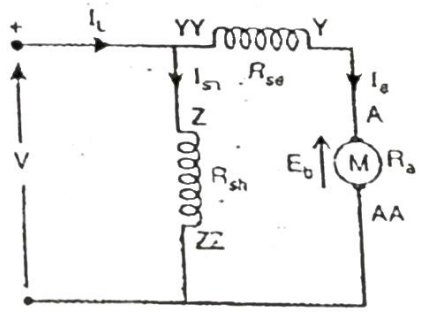
The series field winding carries the armature current and should have very small resistance so that the voltage drop across it is very small. Hence, it is made of a few thick turns of copper. The back E.M.F. induced in the armature is given by,

$$E_b = V - I_a (R_a + R_{se}) - \text{B.C.D.} - \text{A.R.D.} \qquad (8.22)$$

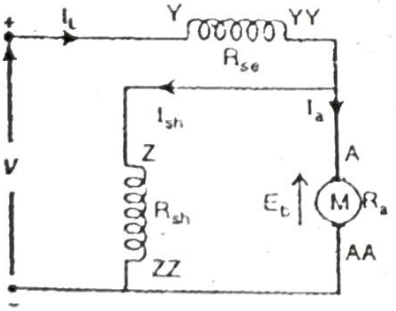
8.9 D.C. Compound Motor:



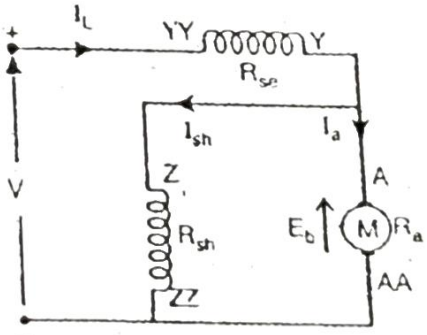
a) Cumulatively compounded long shunt DC motor



b) Differentially compounded long shunt DC motor



c) Cumulatively compounded short shunt DC motor



d) Differentially compounded short shunt DC motor

Fig.8.6

A D.C. compound motor contains both shunt field winding and series field winding. If the fluxes, ϕ_{sh} produced by the shunt field winding and ϕ_{se} produced by the series field winding are in the same direction and are additive, then the motor is said to be *cumulatively compounded*. If the two fluxes oppose each other, then the motor is said to be

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differentially compounded. Depending on the way in which the two field windings are connected, the compound motors can be either *long shunt* or *short shunt*. All the four possible types of D.C. compound motors are shown in Figs. 8.6 (a), (b), (c) and (d).

For cumulatively compounded motors, we observe that the currents enter the positive terminals of the two field windings and hence the fluxes produced by them are in the same direction and they are additive. In the case of differentially compounded motors, the current through the series field winding enters the negative terminal and the current through the shunt field winding enters the positive terminal. Hence, the two fluxes produced are in opposite directions and hence they oppose each other.

For a cumulatively or differentially compounded long shunt, D.C. motor, the following equations hold good.

$$I_{sh} = \frac{V}{R_{sh}} \qquad (8.23) \qquad I_a = I_L - I_{sh} \qquad (8.24)$$

And $E_b = V - I_a (R_a + R_{se}) - B.C.D. - A.R.D.$ (8.25)

For a cumulatively or differentially compounded short shunt, D.C. motor, the following equations hold good.

$$I_{sh} = \frac{V - I_L R_{se}}{R_{sh}} \qquad (8.26) \qquad I_a = I_L - I_{sh} \qquad (8.27)$$

And $E_b = V - I_L R_{se} - I_a R_a - B.C.D - A.R.D.$ (8.28)

8.10 Characteristics of D.C. Motors:

The characteristics of D.C. motors are studied keeping the applied voltage constant. The following are the three important characteristics of D.C. motors.

i) Electrical characteristic or T_a / I_a characteristic
ii) N / I_a characteristic and
iii) Mechanical Characteristic or N / T_a characteristic.

8.11 Characteristics of D.C. Shunt Motors:

(i) T_a / I_a Characteristic:

Fig. 8.7 represents a D.C. shunt motor on load. In a D.C. shunt motor, the field current I_{sh} remains constant irrespective of the load connected to the motor, because, the applied voltage remains constant for all loads. Hence the flux produced also remains constant.

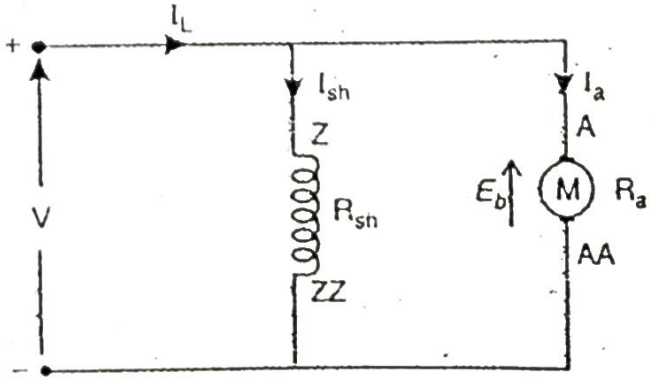


Fig.8.7

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$$T_a = 0.159 \phi Z I_a \left(\frac{P}{A}\right)$$

In the above equation, Z, P, A and ϕ are constants. $\therefore$ $T_a \alpha I_a$ (8.29)

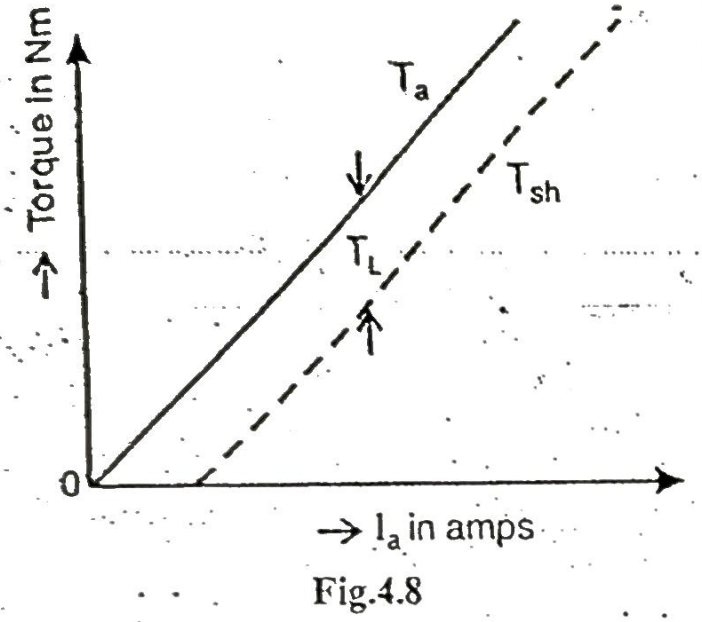


Fig.4.8

Hence T_a / I_a characteristic is a straight line passing through the origin as shown fig. 8.8.

The shaft torque T_{sh} is always less than the armature torque T_a due to iron losses and mechanical losses in the D.C. motor. Hence it is shown always to be less than T_a by a torque T_L, which is the torque lost due to the above losses. From the characteristic, we learn that, a D.C. shunt motor has a medium starting torque and hence does not suit where, very large loads are required to be started.

(ii) N / I_a Characteristic:

The equation for the back E.M.F. E_b is given by,

$$E_b = \frac{\phi Z N P}{60 A} \quad \therefore \quad N \alpha \frac{E_b}{\phi} \qquad (8.30)$$

as the other quantities are constant.

For D.C. shunt motor, ϕ is constant

$$\therefore \quad N \alpha E_b \alpha V - I_a R_a \qquad (8.31)$$

As I_a increases, $I_a R_a$ increases and hence the speed decreases. But the drop $I_a R_a$ is very small compared to V and hence, the decrease of speed as the armature current increases is also small. The variation of speed with respect to armature current is as shown in Fig.8.9.

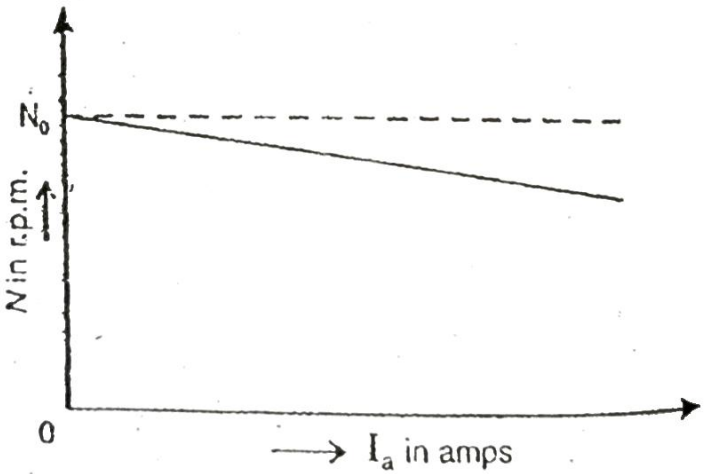


Fig.8.9

N_0 is the no load speed of the motor. From this characteristic, we understand that the change in the speed of a D.C. shunt motor is small, when the armature current or load on the motor is increased. Hence for all practical purposes, a D.C. shunt motor may be considered as almost a constant speed motor.

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iii) N / T_a Characteristic:

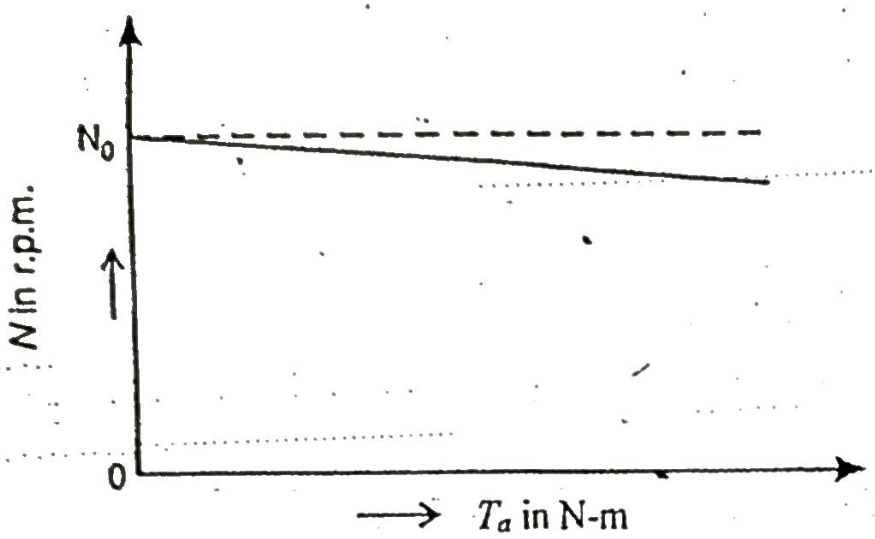


Fig.8.10

From the equation (8.29), we know that $T_a \propto I_a$ and hence, N/T_a characteristic is similar to N/I_a characteristic as shown in fig.8.10. N_0 is the no load speed of the motor.

8.12 Characteristics of D.C. Series Motors:

(a) T_a / I_a Characteristic:

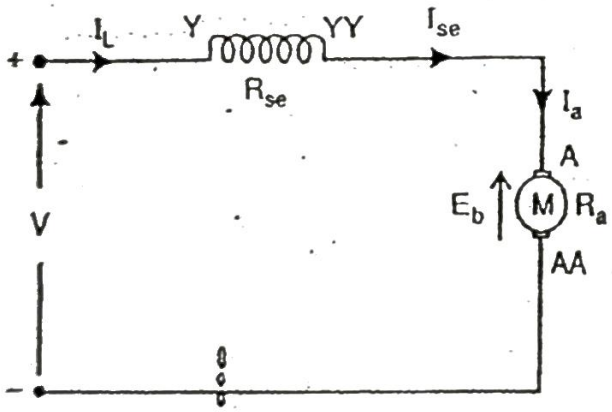


Fig.8.11

Fig. 8.11 represents a D.C. series motor on load. As the load on the motor increases, the current through the series field winding also increases and hence the flux produced also increases. The torque equation of a D.C. motor is given by,

$$T_a = 0.159 \phi Z I_a \left(\frac{P}{A}\right)$$

Or $T_a \propto \phi I_a$, but $\phi \propto I_a$, $\therefore T_a \propto I_a^2$

But after saturation, the flux remains constant and $T_a \propto I_a$.

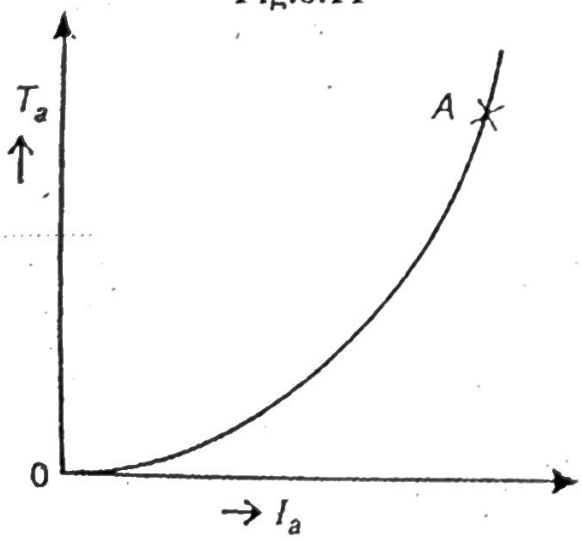


Fig.8.12

The variation of T_a with respect to I_a is as shown in Fig. 8.12.

Upto saturation i.e., upto point A, $T_a \propto I_a^2$ and hence the curve is a parabola. After saturation i.e., beyond point A, $T_a \propto I_a$ and hence the curve is a straight line. From the characteristic, we find that the starting torque of a D.C. series motor is very high.

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(b) N / I_a Characteristic:

From equation (8.30), we know that, $N \alpha \frac{E_b}{\phi} \alpha \frac{V - I_a(R_a + R_{se})}{\phi}$ (8.31)

From the equation (8.31), we find that as the load on the motor increases, there are two factors which influence the speed of the motor.

(i) $I_a (R_a + R_{se})$ increases and hence the speed decreases.
(ii) The flux ϕ also increases due to which the speed decreases.

But it has been observed that the decrease of speed due to the first factor is negligibly small as compared to the decrease in speed due to the second factor. Hence for all practical purpose, we can say

$$N \alpha \frac{1}{\phi} \quad \text{but} \quad \phi \alpha I_a, \quad \therefore \quad N \alpha \frac{1}{I_a} \qquad (8.32)$$

The variation of speed with respect to I_a is as shown in fig. 8.13.

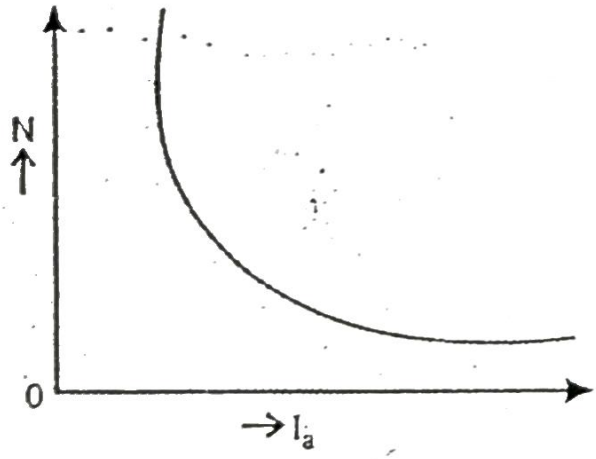


Fig.8.13

From the characteristic, we observe that, as the load increases, the speed decreases over a wide range. Hence, a D.C. series motor is considered as a variable speed motor.

At no load, I_a is very small and hence the speed will be dangerously high as per equation (8.32). Hence, if a D.C. series motor is started without any load on it, the speed is very high and it may run out of the foundation due to the centrifugal forces set up. Hence, a D.C. series motor should never be started without load.

(c) N / T_a Characteristic:

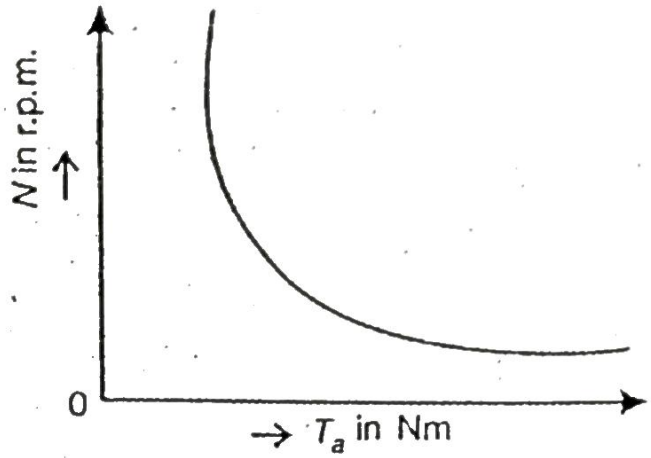


Fig.8.14

From the N / I_a characteristic of a D.C. series motor shown in fig.8.13, we find that, when I_a is small, N is very high.

But, $T_a \alpha I_a^2$ or $I_a \alpha \sqrt{T_a}$

$$\therefore \quad N \alpha \frac{1}{\sqrt{T_a}} \qquad (8.33)$$

For smaller values of T_a, N is very large and for higher value of T_a the speed decreases. The variation of speed with respect to torque T_a is as shown in Fig.8.14

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8.13 Characteristics of D.C. Compound Motors:

(a) T_a / I_a Characteristic:

In the case of a cumulatively compounded D.C. motor, as the load increases, I_a also increases. As the current I_a increases, the flux ϕ_{se} produced by the series field also increases but as the shunt field current I_{sh} remains constant, the flux ϕ_{sh} produced by it also remains constant. However the total flux ϕ which is the sum of ϕ_{se} and ϕ_{sh} increases.

We know that, $T_a \alpha \phi I_a$

Hence, the torque produced also increases, but at a faster rate than in the case of D.C. shunt motor, and at a lesser rate than in the case of D.C. series motor. The variation of T_a with respect to I_a is shown in Fig. 8.15.

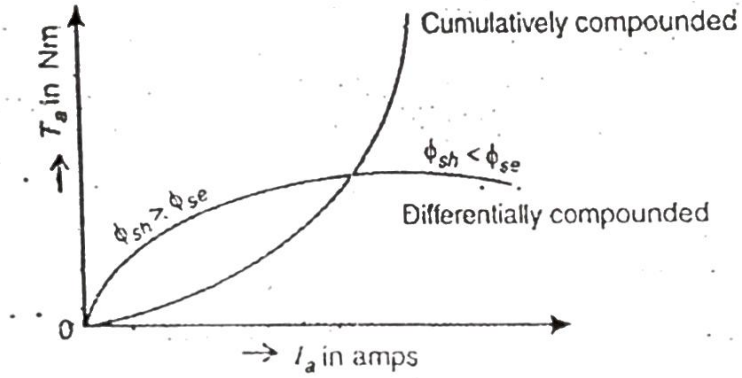


Fig.8.15

A cumulatively compounded D.C. motor has a starting torque more than that of a D.C. shunt motor and less than that of a D.C. series motor.

In the case of a differentially compounded D.C. motor, as the load increases, I_a increases. As I_a flowing through the series field winding increases, the flux ϕ_{se} produced by it also increases. The flux ϕ_{sh} produced by the shunt field winding remains constant. ϕ_{se} opposes ϕ_{sh} and hence the resultant flux is given by,

$$\phi = \phi_{sh} - \phi_{se}$$

As the armature torque $T_a \alpha \phi I_a$, the increase of I_a increases the torque T_a, but the decrease of ϕ decreases the torque T_a.

As long as $\phi_{sh} > \phi_{se}$, the torque T_a increases with I_a, but once $\phi_{se} > \phi_{sh}$ the torque T_a starts decreasing with the increase of I_a. The variation of T_a with respect to I_a is as shown in Fig. 8.15. The decrease of torque with the increase of load is an undesirable characteristic.

(b) N / I_a Characteristic:

We know that, for a D.C. motor, $$N \alpha \frac{E_b}{\phi} \alpha \frac{V - I_a R_a}{\phi} \qquad (8.34)$$

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$E_{b1} = V - I_a R_a = 250 - 20 \times 0.5 = 240V$

Let R be the resistance connected in series with the armature. As the torque remains the same, the armature current also remains the same.

$E_{b2} = V - I_a (R_a + R) = 250 - 20 \times (0.5 + R) = 240 - 20\,R$

$\frac{E_{b1}}{E_{b2}} = \frac{N_1}{N_2}$ i.e. $\frac{240}{240 - 20R} = \frac{1000}{500}$, $\therefore \quad R = 6\,\Omega$

8.13 A 240 V, 4 pole shunt motor running at 1,000 r.p.m. gives 15 H.P. with an armature current of 50A and a field current of 1.0A. The armature winding is wave connected and has 540 conductors. Its resistance is 0.1 Ω and the drop at each brush is 1V. Find (a) useful torque. (b) total torque (c) useful flux per pole (d) rotational losses and (e) efficiency.

Output = 15 x 735.5 = 11,032.5 watts, $I_L = I_a + I_{sh} = 50 + 1 = 51A$

Input = V I_L = 240 x 51 = 12,240 watts, $\therefore \quad \eta = \frac{11032.5}{12240} \times 100 = 90\%$

$E_b = V - I_a R_a - B.C.D. = 240 - 50 \times 0.1 - 2 \times 1 = 233V$

Armature output = $E_b I_a = 233 \times 50 = 11{,}650$ watts

$$\therefore T_a = \frac{\text{Armature output in watts}}{2\pi \frac{N}{60}} = \frac{11{,}650}{2 \times 3.14 \times \frac{1000}{60}} = 111.31 \text{ Nm}$$

The useful torque is given by

$$T_{sh} = \frac{\text{Motor output in watts}}{2\pi \frac{N}{60}} = \frac{11032.5}{2 \times 3.14 \times \frac{1000}{60}} = 105.41 \text{ Nm}$$

We know, $E_b = \frac{\phi Z N P}{60A}$, $\therefore 233 = \frac{\phi \times 540 \times 1000 \times 4}{60 \times 2}$

$\therefore \phi = 0.0129$ Wb or 12.9 mWb

Lost torque = $T_L = T_a - T_{sh} = 111.31 - 105.41 = 5.9$ Nm

$$T_L = \frac{\text{Rotational losses in watts}}{2\pi \frac{N}{60}} = \frac{\text{Rotational losses in watts}}{2 \times 3.14 \times \frac{1000}{60}} = 5.9$$

$\therefore$ Rational losses = 617.53 watts

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8.14 Applications of D.C. Motors:

(a) D.C. Shunt Motors:

From the study of the characteristics of a D.C. shunt motor, we realize that it has a medium starting torque and its speed remains almost constant from no load to full load. Hence, it is used where constant speed is required and the starting torque required is not very high, such as for lathes, centrifugal pumps, fans, reciprocating pumps, drilling machines, boring machines, spinning and weaving machines etc.,

(b) D.C. Series Motors:

From the study of the characteristics of D.C. series motor, we realize that, it has a very high starting torque and its speed varies widely from no load to full load. Hence, it is used where very high starting torque and variable speed is required, such as for electric traction work, electric locomotives, trolleys, cranes, hoists, conveyors, air compressors, vacuum cleaners, hair driers, sewing machines etc.,

(c) D.C. Compound Motors:

The characteristics of cumulatively compounded D.C. motor is between those of D.C. shunt motor and D.C. series motor. It has high starting torque and a variable speed. It is used where sudden loads are applied or removed such as for shears, punches, coal cutting machines, elevators, conveyors, heavy planers, rolling mills, ice machines, printing presses, air compressors etc. The differentially compounded D.C. motor has no practical applications, because of its undesirable characteristics.

8.15 Necessity of a Starter for a D.C. Motor:

For a D.C. motor, we know that the armature current is given by,

$$I_a = \frac{V - E_b}{R_a} \qquad (8.36)$$

When the motor is at rest, $E_b = 0$, $\therefore \quad I_a = \frac{V}{R_a}$

The armature resistance R_a is very small and is usually of the order of 0.5 Ω, for a 5 HP, 220V motor. If a voltage of 220V is applied to the motor, the starting current that flows through the armature is given by,

$$I_a = \frac{220}{0.5} = 440 \text{ A}$$

This current is very large compared to the rated current of the motor which is given by,

$$I_{f.l} = \frac{5 \times 735.5}{220} = 16.72 \text{ A}$$

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The large current of 440 A that flows through the armature will blow out the fuses or damage the commutator, brushes and the armature winding. Hence, a starter is necessary for a D.C. motor.

A starter for a D.C. motor is nothing but a resistance connected in series with the armature during starting and which is gradually cut out, so that, the motor picks up its rated speed. When the motor is running, back E.M.F. is developed and the armature current will be within the limits and the starting resistance is not required.

8.16 Three Point Starter:

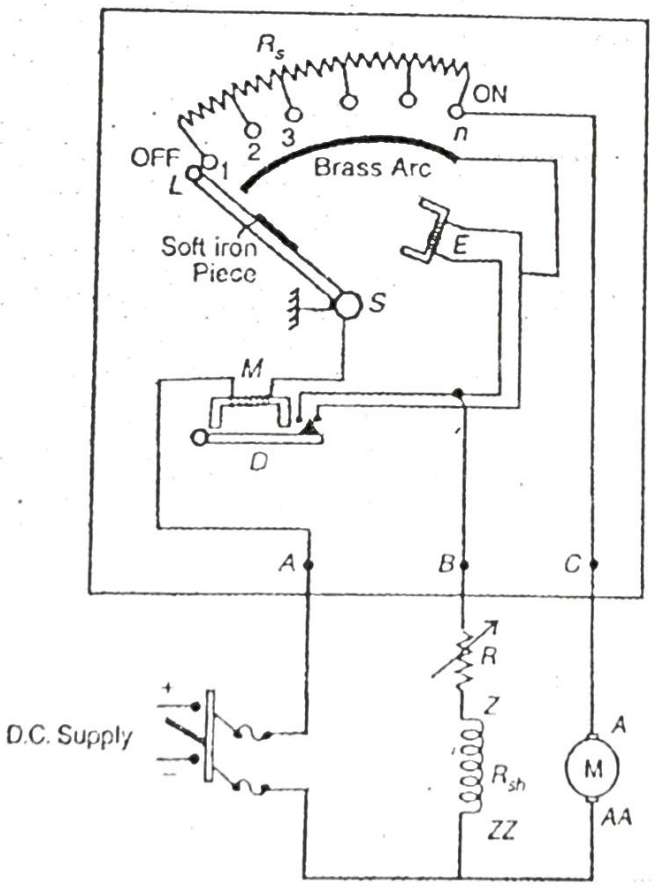


Fig.8.18

A three point starter with its internal wiring is shown in Fig. 8.18. A, B and C are the three terminals of the starter, to which external connections are made. R_S is the starting resistance, which is divided into various sections of equal resistances and these sections are connected to brass studs 1, 2, 3... n. The +ve line of the supply is connected to the point A. The field winding is connected to point B, along with a variable resistance R, which is used to control the speed of the motor. The +ve terminal A of the armature is connected to point C.

To start the motor, the supply switch is closed and the starting brass arm L is moved to the right. As soon as the brass arm touches the stud number 1 of the starting resistance Rs, it also touches the brass arc, thus giving supply to the shunt field winding as well as to the

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armature. The motor starts rotating. The starting arm is moved gradually on the remaining studs till the starting resistance is completely cut out. The speed of the motor gradually increases and almost reaches the rated speed. If the rated speed is not reached, then the resistance R which is in series with the shunt field winding is slowly varied to bring the motor to the rated speed. As long as the motor is running and the supply is on, the brass arm L is held in the ON position by the electromagnet E, which keeps the soft iron piece which is attached to the arm, attracted to it. When the motor is running at its rated speed, the starting resistance is completely out of the armature circuit.

There are two protective devices in the starter. One is the electromagnet E, which is also known as the HOLD ON COIL or NO VOLTAGE RELEASE. When the power fails and the motor is in running condition, the electromagnet E is de-energised and the spring S attached to the other end of the brass arm, pulls it back to OFF position. This prevents the stationary motor from being put across the supply without the starting resistance, when the power supply is resumed again. The electromagnet E also prevents the motor from reaching dangerously high speed, when the field circuit is opened by chance, when the motor is running.

The other protective device is the electromagnet M, which is known as the OVER LOAD RELEASE. As long as the motor is drawing rated current and less, the arm D is not attracted by M. But when the current increases beyond the rated value, M attracts D, thereby short circuiting the electromagnet E. The electromagnet E gets de-energised and hence the arm L is pulled back to OFF position.

The three point starter is usually used to start D.C. shunt and D.C. compound motors.

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WORKED EXAMPLES

8.1 **Determine the total torque developed in a 250 V, 4 pole D.C. shunt motor with lap winding accommodated in 60 slots, each containing 20 conductors. The armature current is 50 A and the flux per pole is 23 mWb.**

$$T_a = 0.159\,\phi\,Z\,I_a\left(\frac{P}{A}\right) = 0.159 \times 23 \times 10^{-3} \times (60 \times 20) \times 50 \times \frac{4}{4} = 219.42 \text{ Nm}$$

8.2 **A 4 pole, 220V, lap connected D.C. shunt motor has 36 slots, each slot containing 16 conductors. It draws a current of 40 A from the supply. The field resistance and armature resistances are 110 Ω and 0.1 Ω respectively. The motor develops an output power of 6 kW. The flux per pole is 40 mWb. Calculate (a) the speed (b) the torque developed by the armature and (c) the shaft torque.**

$$I_{sh} = \frac{V}{R_{sh}} = \frac{220}{110} = 2\text{A}$$

$$I_a = I_L - I_{sh} = 40 - 2 = 38 \text{ A}$$

$$E_b = V - I_a R_a = 220 - 38 \times 0.1 = 216.2 \text{ V}$$

$$E_b = \frac{\phi Z N P}{60A}$$

$$\text{i.e. } 216.2 = \frac{40 \times 10^{-3}(36 \times 16) \times N \times 4}{60 \times 4}$$

$$\therefore N = 563 \text{ r.p.m}$$

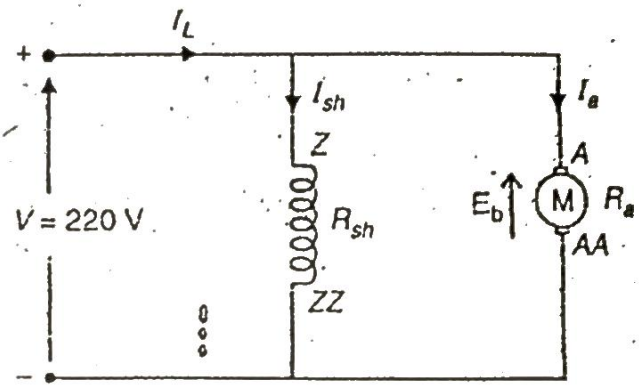


Fig.8.19

$$T_a = \frac{E_b I_a}{2\pi \frac{N}{60}} = \frac{216.2 \times 38}{2 \times 3.14 \times \frac{563}{60}} = 139.42 \text{ Nm}$$

$$T_{sh} = \frac{6 \times 1000}{2 \times 3.14 \times \frac{563}{60}} = 101.82 \text{ Nm}$$

8.3 **The armature current of a series motor is 60 A, when on full load. If the load is adjusted so that this current decreases to 40 A, find the new torque expressed as a percentage of full load torque. The flux for a current of 40 A is 70% of that when the current is 60A.**

$$T_{f\ell} = 0.159\,\phi\,Z\,I_a\left(\frac{P}{A}\right) = 0.159 \times \phi \times Z \times 60\left(\frac{P}{A}\right)$$

$$T_{40A} = 0.159 \times (0.7\phi) \times Z \times (40)\left(\frac{P}{A}\right)$$

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WORKED EXAMPLES

8.1 Determine the total torque developed in a 250 V, 4 pole D.C. shunt motor with lap winding accommodated in 60 slots, each containing 20 conductors. The armature current is 50 A and the flux per pole is 23 mWb.

$$T_a = 0.159\,\phi\, Z\, I_a \left(\frac{P}{A}\right) = 0.159 \times 23 \times 10^{-3} \times (60 \times 20) \times 50 \times \frac{4}{4} = 219.42 \text{ Nm}$$

8.2 A 4 pole, 220V, lap connected D.C. shunt motor has 36 slots, each slot containing 16 conductors. It draws a current of 40 A from the supply. The field resistance and armature resistances are 110 Ω and 0.1 Ω respectively. The motor develops an output power of 6 kW. The flux per pole is 40 mWb. Calculate (a) the speed (b) the torque developed by the armature and (c) the shaft torque.

$$I_{sh} = \frac{V}{R_{sh}} = \frac{220}{110} = 2\text{A}$$

$$I_a = I_L - I_{sh} = 40 - 2 = 38 \text{ A}$$

$$E_b = V - I_a R_a = 220 - 38 \times 0.1 = 216.2 \text{ V}$$

$$E_b = \frac{\phi Z N P}{60A}$$

i.e. $$216.2 = \frac{40 \times 10^{-3}(36 \times 16) \times N \times 4}{60 \times 4}$$

$$\therefore N = 563 \text{ r.p.m}$$

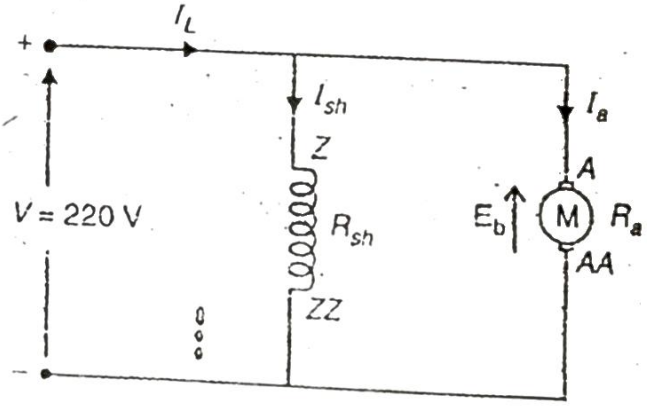


Fig.8.19

$$T_a = \frac{E_b I_a}{2\pi \frac{N}{60}} = \frac{216.2 \times 38}{2 \times 3.14 \times \frac{563}{60}} = 139.42 \text{ Nm}$$

$$T_{sh} = \frac{6 \times 1000}{2 \times 3.14 \times \frac{563}{60}} = 101.82 \text{ Nm}$$

8.3 The armature current of a series motor is 60 A, when on full load. If the load is adjusted so that this current decreases to 40 A, find the new torque expressed as a percentage of full load torque. The flux for a current of 40 A is 70% of that when the current is 60A.

$$T_{fl} = 0.159\,\phi\, Z\, I_a \left(\frac{P}{A}\right) = 0.159 \times \phi \times Z \times 60 \left(\frac{P}{A}\right)$$

$$T_{40A} = 0.159 \times (0.7\phi) \times Z \times (40) \left(\frac{P}{A}\right)$$

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$$\therefore \quad \frac{T_{40A}}{T_{f.l.}} = \frac{0.7 \times 40}{60} = 0.4667 \quad \text{or} \quad 46.67\,\%$$

$$\therefore \quad T_{40A} = 0.4667\ T_{f.l.}$$

8.4 **A shunt D.C. machine connected to 250 V supply, has an armature resistance of 0.12 Ω and the field resistance of 100 Ω. Find the ratio of the speed of the machine as a generator to the speed as a motor, the line current in each case being 80 A.**

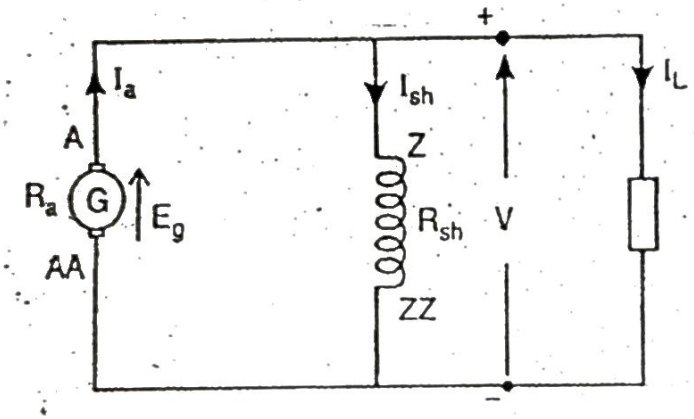


Fig.8.20

Machine as a generator

$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{100} = 2.5\text{A}$$

$$I_a = I_L + I_{sh} = 80 + 2.5 = 82.5\ \text{A}$$

$$E_g = V + I_a R_a = 250 + 82.5 \times 0.12$$

$$= 259.9\ \text{V}$$

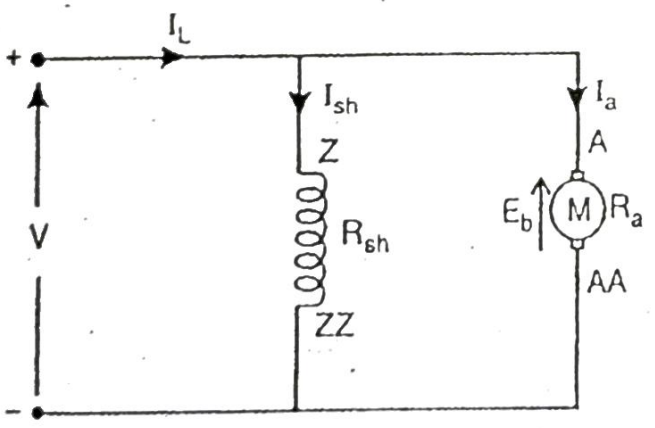


Fig.8.21

Machine as a motor

$$I_a = I_L - I_{sh}$$

$$= 80 - 2.5 = 77.5\ \text{A}$$

$$E_b = V - I_a R_a$$

$$= 250 - 77.5 \times 0.12$$

$$= 240.7\ \text{V}$$

$$\therefore \quad \frac{N_g}{N_m} = \frac{E_g}{E_b} = \frac{259.9}{240.7} = 1.08$$

8.5 **A 4 pole, 500 V shunt motor has 720 wave connected conductors on its armature. The full load armature current is 60 A and the flux per pole is 0.03Wb. The armature resistance is 0.2 Ω and the contact drop is 1 V per brush. Calculate the full load speed.**

$$E_b = V - I_a R_a - \text{B.C. D} = 500 - 60 \times 0.2 - 2 \times 1 = 486\ \text{V}$$

$$\text{But } E_b = \frac{\phi Z N P}{60A} \quad \text{i.e.} \quad 486 = \frac{0.03 \times 720 \times N \times 4}{60 \times 2}, \quad \therefore \quad N = 675\ \text{r.p.m.}$$

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8.6 The armature resistance of a 25 H.P., 250V D.C. series motor is 0.2 Ω and the series field resistance is 0.05 Ω. When the series motor takes 80 A, the speed is 600 r.p.m. Calculate the speed when the current is 50 A.

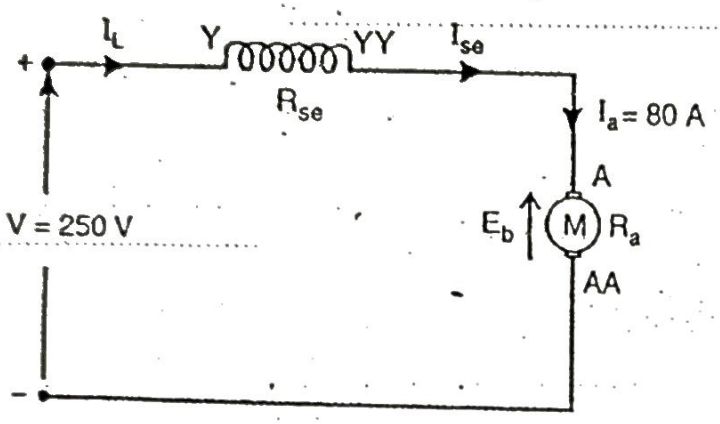


Fig.8.22

$$E_{b1} = V - I_{a1}(R_a + R_{se})$$
$$= 250 - 80 \times (0.05 + 0.2)$$
$$= 230 \text{ V}$$

$$E_{b2} = V - I_{a2}(R_a + R_{se})$$
$$= 250 - 50 \times (0.05 + 0.2)$$
$$= 237.5 \text{ V}$$

$$N_1 \alpha \frac{E_{b1}}{\phi_1} \alpha \frac{E_{b1}}{I_{a1}}, \qquad N_2 \alpha \frac{E_{b2}}{\phi_2} \alpha \frac{E_{b2}}{I_{a2}}$$

$$\therefore \frac{N_1}{N_2} = \frac{E_{b1}}{I_{a1}} \times \frac{I_{a2}}{E_{b2}} \quad \text{i.e.} \quad \frac{600}{N_2} = \frac{230}{80} \times \frac{50}{237.5}, \quad \therefore \quad N_2 = 991.3 \text{ r.p.m}$$

8.7 A 250V, D.C. shunt motor takes 6 A line current on no load and runs at 1000 r.p.m. The resistance of the field winding and armature are 250 Ω and 0.2 Ω respectively. If the full load line current is 26 A, Calculate the full load speed, assuming constant air gap flux.

$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{250} = 1\text{A}$$

$$I_{a0} = 6 - 1 = 5\text{A}$$

$$E_{b0} = V - I_{a0} R_a$$
$$= 250 - 5 \times 0.2$$
$$= 249 \text{ V}$$

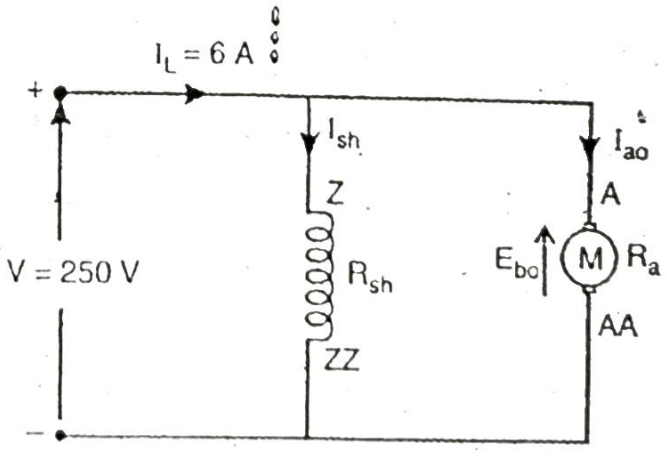


Fig.8.23

On full load:

$I_{a1} = 26 - 1 = 25$ A, $\quad E_{b1} = 250 - 25 \times 0.2 = 245$ V

$$\frac{N_0}{N_1} = \frac{E_{b0}}{E_{b1}} \quad \text{i.e.} \quad \frac{1000}{N_1} = \frac{249}{245}, \quad \therefore \quad N_1 = 984 \text{ r.p.m.}$$

8.8 A 250V shunt motor on no load runs at 1000 r.p.m. and takes 5A. The total armature and shunt field resistances are 0.2 Ω and 250 Ω respectively. Calculate the speed when loaded and taking a current of 50A, if the armature reaction weakens the field by 3 %.

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On no load

$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{250} = 1A$$

$$I_{a0} = I_{L0} - I_{sh} = 5 - 1 = 4A$$

$$E_{b0} = 250 - 4 \times 0.2 = 249.2\ V$$

When loaded

$$I_L = 50\ A$$

$$I_{a1} = 50 - 1 = 49\ A$$

$$E_{b1} = 250 - 49 \times 0.2 = 240.2\ V$$

We know, $E_b \propto N\phi$

$\therefore E_{b0} \propto N_0 \phi_0$ and $E_{b1} \propto N_1 \phi_1$

$$\frac{E_{b1}}{E_{b0}} = \frac{N_1 \phi_1}{N_0 \phi_0} = \frac{N_1}{N_0} \times 0.97$$

i.e. $$\frac{240.2}{249.2} = \frac{N_1}{1000} \times 0.97$$

$\therefore N_1 = 993.7$ r.p.m.

8.9 Find the useful flux per pole on no load of a 250 V, 6 pole shunt motor having a 2 circuit, wave connected armature winding with 110 turns. At normal working temperature, the overall armature resistance including brushes is 0.2 Ω. The armature current is 13.3 A at the no load speed of 908 r.p.m.

$E_{b0} = V - I_a R_a = 250 - 13.3 \times 0.2 = 247.34$ V

$$E_{b0} = \frac{\phi ZNP}{60A} \quad \text{i.e.} \quad 247.34 = \frac{\phi \times (110 \times 2) \times 908 \times 6}{60 \times 2}$$

$\therefore \phi = 0.02476$ Wb $= 24.76$ mWb

88.10 A 240 V, unsaturated shunt motor has an armature resistance of 0.04 Ω and field resistance of 100 Ω. (a) Find what resistance must be added to the field circuit to increase the speed from 1,200 to 1,500 r.p.m. when the supply current is 200A.

$$I_{sh1} = \frac{240}{100} = 2.4\ A$$

$I_{a1} = I_L - I_{sh1} = 200 - 2.4$

$= 197.6$ A

$E_{b1} = V - I_{a1} R_a$

$= 240 - 197.6 \times 0.04$

$= 232.096$ V

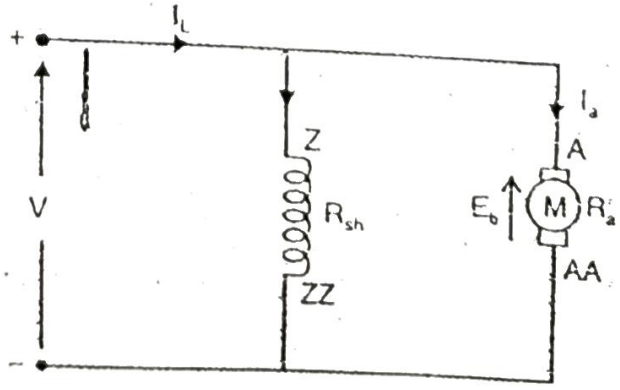


Fig.8.24

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Let R be the resistance connected in series with the field resistance. Then

$$I_{sh2} = \frac{240}{(100+R)}, \qquad I_{a2} = 200 - \frac{240}{100+R}$$

$$E_{b2} = 240 - \left(200 - \frac{240}{100+R}\right)0.04$$

$$\frac{E_{b1}}{E_{b2}} = \frac{N_1\phi_1}{N_2\phi_2} = \frac{N_1 I_{sh1}}{N_2 I_{sh2}}$$

$$\therefore \frac{232.096}{240 - \left(200 - \frac{240}{100+R}\right)0.04} = \frac{1200}{1500} \times \frac{2.4}{\frac{240}{100+R}}$$

On solving, R = 12.5 Ω

8.11 A 100 kW, belt-driven shunt generator running at 300 r.p.m. on 220 V bus bars continues to run as a motor when the belt breaks, then taking 10 kW. What will be its speed? Armature resistance = 0.025 Ω, field resistance is 60 Ω. Contact drop under each brush = 1 V. Ignore armature reaction.

As a generator, $I_{sh} = \frac{220}{60} = 3.67$ A, $I_{L1} = \frac{P}{V} = \frac{100\times10^3}{220} = 454.55$A

$I_{a1} = I_{L1} + I_{sh} = 454.55 + 3.67 = 458.22$ A

$E_g = V + I_{a1}R_a + B.C.D. = 220 + 458.22 \times 0.025 + 2 \times 1 = 233.45$V

When the belt breaks, generator continues to run as a motor.

$I_{L2} = \frac{P}{V} = \frac{10\times10^3}{220} = 45.46$ A, $I_{a2} = I_{L2} - I_{sh} = 45.46 - 3.67 = 41.79$ A

$E_b = V - I_{a2}R_a - B.C.D. = 220 - 41.79 \times 0.025 - 2 \times 1 = 216.95$ V

We know, $\frac{E_g}{E_b} = \frac{N_1}{N_2} = \frac{233.45}{216.95} = \frac{300}{N_2}$

$\therefore N_2 = 278.79$ r.p.m. $\cong 279$ r.p.m.

8.12 A 250 V shunt motor has an armature current of 20 A when running at 1000 r.p.m. against full load torque. The armature resistance is 0.5 Ω. What resistance must be inserted is series with the armature to reduce the speed to 500 r.p.m. at the same torque and what will be the speed, if the load torque is halved with this resistance in circuit? Assume the flux to remain constant throughout.

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8.14 A 220 V, shunt motor with an armature resistance of 0.5 Ω is excited to give a constant main field. At full load, the motor runs at 500 r.p.m. and takes an armature current of 30 A. If a resistance of 1 Ω is placed in the armature circuit, find the speed at (a) Full load torque (b) Double the full load torque and (c) the stalling torque.

(a) At full load, $E_{b1} = V - I_{a1} R_a = 220 - 30 \times 0.5 = 205$ V

When a resistance of 1 Ω is placed in the armature circuit.

$E_{b2} = V - I_{a1}(R_a + 1) = 220 - 30 \times (0.5 + 1) = 175$ V

We know, $\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}}$ i.e. $\frac{500}{N_2} = \frac{205}{175}$

$\therefore N_2 = 426.83$ r.p.m.

Hence the speed at full load torque = 426.83 r.p.m. ≅ 427 r.p.m.

(b) At double the full load torque, $I_{a2} = 30 \times 2 = 60$A

With resistance 1 Ω in the armature circuit

$E_{b3} = V - I_{a2}(R_a + 1) = 220 - 60 \times (0.5 + 1) = 130$V

We know, $\frac{E_{b2}}{E_{b3}} = \frac{N_2}{N_3}$ i.e. $\frac{175}{130} = \frac{426.83}{N_3}$, $\therefore N_3 = 317.07$ r.p.m.

Hence, the speed at double the full load torque = 317.07 r.p.m. ≅ 317 r.p.m.

(c) Stalling torque is nothing but the starting torque at which the speed is zero.

$N_4 = 0$, $E_{b4} = 0$

The starting armature current is calculated using the formula

$E_{b4} = V - I_{a4}(R_a + 1) = 220 - I_{a4}(0.5 + 1) = 0$

$\therefore I_{a4} = 146.67$ A

As the flux is constant, $T \propto I_a$

$$\therefore \frac{T_{st}}{T_{f.l}} = \frac{\text{Starting current } I_{a4}}{\text{Full load current } I_{a1}} = \frac{146.67}{30} = 4.889$$

8.15 A 200 V series motor which has a total resistance of 0.2 Ω runs at 1000 r.p.m. and takes full load current of 200 A. Calculate the speed of the motor at which it develops half the full load torque.

$E_{b1} = 200 - 200 \times 0.2 = 160$V

We know that, $T \propto \phi I_a$

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$T_1 \alpha \phi_1 I_{a1} \alpha I_{a1}^2$ $\quad \because \phi_1 \alpha I_{a1}$

$T_2 \alpha \phi_2 I_{a2} \alpha I_{a2}^2$ $\quad \because \phi_2 \alpha I_{a2}$

$\frac{T_1}{T_2} = \frac{I_{a1}^2}{I_{a2}^2} = 2, \quad \therefore I_{a2} = \sqrt{\frac{I_{a1}^2}{2}} = \sqrt{\frac{200^2}{2}} = 141.42 \text{ A}$

$E_{b2} = 200 - 141.42 \times 0.2 = 171.716 \text{ V}$

We know that $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2} = \frac{E_{b2}}{E_{b1}} \times \frac{I_{a1}}{I_{a2}}$

i.e. $\frac{N_2}{1000} = \frac{171.716}{160} \times \frac{200}{141.42}, \quad \therefore N_2 = 1,517.79 \text{ rpm} \cong 1,518 \text{ r.p.m.}$

8.16 A D.C shunt motor takes an armature current of 110 A at 480V. The resistance of the armature circuit is 0.2 Ω. The machine has 6 poles and armature is lap connected with 864 conductors. The flux per pole is 0.05 Wb. Calculate (i) the speed (ii) the gross torque developed by the armature

$E_b = V - I_a R_a \ldots$

$= 480 - 110 \times 0.2 = 458 \text{ V}$

$E_b = \frac{\phi Z N P}{60 A}$

i.e., $458 = \frac{0.05 \times 864 \times N \times 6}{60 \times 6}$

$\therefore N = 636.11 \text{ r.p.m.}$

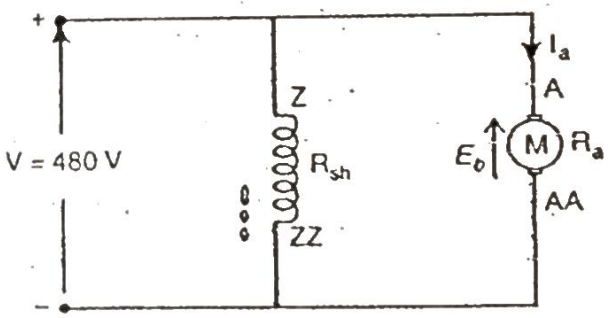


Fig.8.25

$T_a = 0.159 \phi Z I_a \left(\frac{P}{A}\right) = 0.159 \times 0.05 \times 864 \times 110 \times \frac{6}{6} = 755.568 \text{ Nm}$

8.17 A 120 V D.C. shunt motor has an armature resistance of 0.2 Ω and shunt field resistance of 60 Ω. It runs at 1800 r.p.m., when it takes a full load current of 40A. Find the speed of the motor, while it is operating at half the full load, with load terminal voltage remaining the same.

$I_{sh} = \frac{V}{R_{sh}} = \frac{120}{60} = 2\text{A}$

$I_{a1} = 40 - 2 = 38 \text{ A}$

$E_{b1} = V - I_{a1} R_a$

$= 120 - 38 \times 0.2 = 112.4 \text{ V}$

At half full load, $I_L = 20\text{A}$

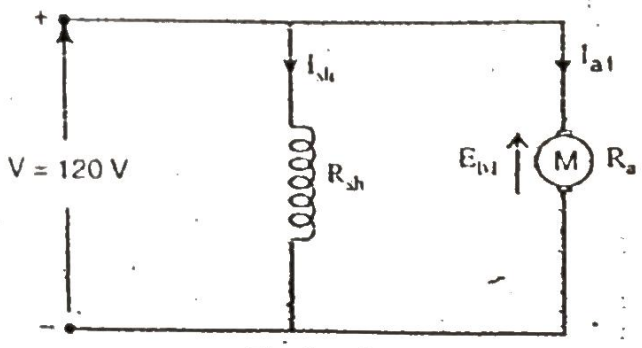


Fig.8.26

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$I_{a2} = 20 - 2 = 18$ A $\qquad E_{b2} = 120 - 18 \times 0.2 = 116.4$ V

$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}} \quad \text{i.e.} \quad \frac{1800}{N_2} = \frac{112.4}{116.4}, \quad \therefore N_2 = 1{,}864.06 \text{ r.p.m.}$$

8.18 4 pole D.C. shunt motor takes 22.5A from a 250 V supply. $R_a = 0.5\ \Omega$ and $R_f = 125\ \Omega$. The armature is wave wound with 300 conductors. If the flux per pole is 0.02 Wb, calculate (a) the speed (b) torque developed and (c) power developed.

$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{125} = 2\text{A}$$

$I_a = I_L - I_{sh} = 22.5 - 2 = 20.5$ A

$E_b = V - I_a R_a = 250 - 20.5 \times 0.5$

$= 239.75$ V

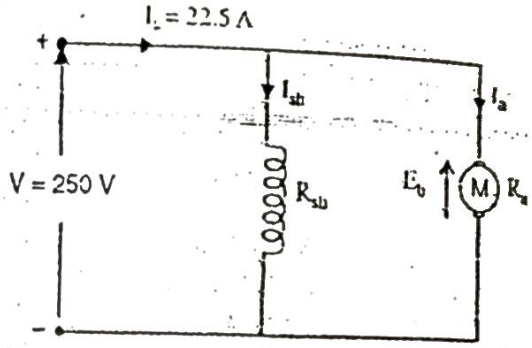


Fig.8.27

$$E_b = \frac{\phi ZNP}{60A} \quad \text{i.e.} \quad 239.75 = \frac{0.02 \times 300 \times N \times 4}{60 \times 2}$$

$\therefore N = 1198.75$ r.p.m

$$T_a = 0.159\,\phi\, Z\, I_a \left(\frac{P}{A}\right) = 0.159 \times 0.02 \times 300 \times 20.5 \times \frac{4}{2} = 39.114 \text{ N-m}$$

$P_a = E_b I_a = 239.75 \times 20.5 = 4914.875$ W

8.19 A series motor runs at 600 r.p.m when taking a current of 110 A from a 230 V supply. The resistance of the armature circuit is 0.12 Ω and that of the series winding is 0.03 Ω. The useful flux per pole for 110 A is 0.024 Wb and that for 50A is 0.0155 Wb. Calculate the speed when the current has fallen to 50A.

$E_{b1} = V - I_{a1}(R_a + R_{se})$

$= 230 - 110\,(0.12 + 0.03) = 213.5$ V

$E_{b2} = 230 - 50\,(0.12 + 0.03) = 222.5$ V

$$N \alpha \frac{E_b}{\phi}, \qquad \therefore \frac{N_1}{N_2} = \frac{E_{b1}}{\phi_1} \times \frac{\phi_2}{E_{b2}}$$

$$\text{i.e.} \quad \frac{600}{N_2} = \frac{213.5}{0.024} \times \frac{0.0155}{222.5},$$

$\therefore N_2 = 968.2$ r.p.m.

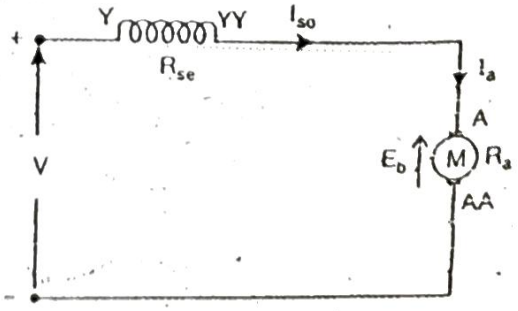


Fig.8.28

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8.20 A 4 pole, 220 V lap connected D.C. shunt motor has 36 slots, each slot containing 16 conductors. It draws a current of 40 A from the supply. The field resistance and armature resistance are 110 Ω and 0.1 Ω respectively. The motor develops an output power of 6 kW. The flux per pole is 40 mWb. Calculate (a) the speed (b) the torque developed by the armature and (c) the shaft torque.

$$I_{sh} = \frac{220}{110} = 2A$$

$$I_a = 40 - 2 = 38\ A$$

$$E_b = 220 - 38 \times 0.1 = 216.2\ V$$

$$216.2 = \frac{40\times10^{-3}\times(36\times16)\times N\times4}{60\times4}$$

$$\therefore N = 563 \text{ r.p.m.}$$

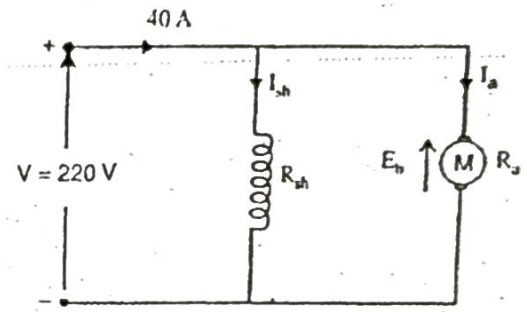


Fig.8.29

$$T_a = \frac{P_a}{2\pi\frac{N}{60}} = \frac{E_b I_a}{2\pi\times\frac{563}{60}}$$

$$= \frac{216.2\times38}{2\pi\times\frac{563}{60}} = 139.42 \text{ N-m}$$

$$T_{sh} = \frac{P_{sh}}{2\pi\frac{N}{60}} = \frac{6\times1000}{2\pi\times\frac{563}{60}} = 101.82 \text{ N-m}$$

8.21 A 4 pole, 250 V, D.C. series motor has wave connected armature with 1254 conductors. The flux per pole is 22 mWb, when the motor is taking 50A. The armature and series field coil resistances are 0.3 Ω and 0.2 Ω respectively. Calculate the speed and torque of the motor and also the power developed in watts.

$$E_b = V - I_a(R_a + R_{se})$$

$$= 250 - 50(0.2 + 0.3) = 225\ V$$

$$E_b = \frac{\phi ZNP}{60A}$$

i.e. $$225 = \frac{22\times10^{-3}\times1{,}254\times N\times4}{60\times2}$$

$$\therefore N = 244.67 \text{ r.p.m.}$$

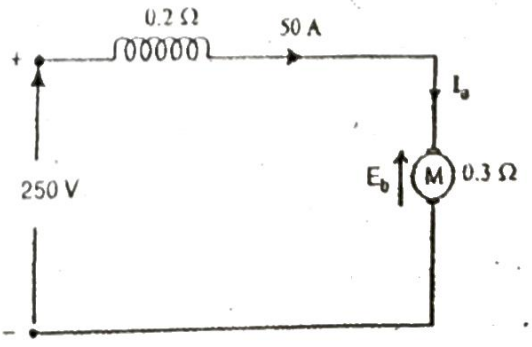


Fig.8.30

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$P_a = E_b I_a = 225 \times 50 = 11{,}250$ W

$P_a = \frac{2\pi N T_a}{60}$ i.e. $11{,}250 = \frac{2\pi \times 244.67 \times T_a}{60}$

$\therefore T_a = 439.08$ N–m

8.22 A 440 V D.C. shunt motor takes an armature current of 20A and runs at 500 r.p.m. The armature resistance is 0.6 Ω. If the flux density is reduced by 30% and the torque is increased by 40%, what are the new values of armature current and speed?

$E_b = V - I_a R_a = 440 - 20 \times 0.6 = 428$ V

$\phi_1 = 0.7\,\phi$ and $T_{a1} = 1.4\,T_a$

$T_a \propto \phi I_a$, $\therefore \frac{T_{a1}}{T_a} = \frac{\phi_1 I_{a1}}{\phi I_a}$

i.e. $\frac{T_{a1}}{T_a} = \frac{0.7\phi \times I_{a1}}{\phi \times 20} = 1.4$,

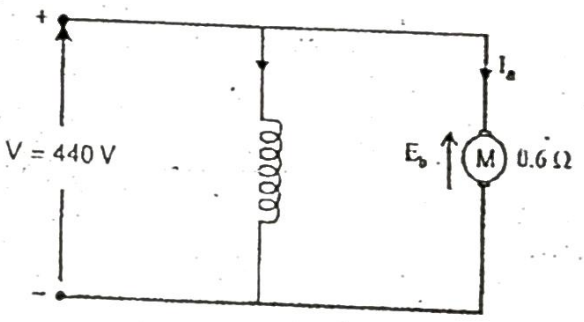


Fig.8.31

$\therefore I_{a1} = 40$A

$E_{b1} = V - I_{a1} R_a = 440 - 40 \times 0.6 = 416$ V

$E_b \propto \phi N$, $\therefore \frac{E_{b1}}{E_b} = \frac{\phi_1 N_1}{\phi N} = \frac{0.7\phi \times N_1}{\phi \times 500}$

i.e. $\frac{416}{428} = \frac{0.7N_1}{500}$, $\therefore N_1 = 694.26$ r.p.m.

8.23 A 6 pole D.C. shunt motor has a lap connected armature with 492 conductors. The resistance of the armature is 0.2 Ω and the flux per pole is 50 mWb. The motor runs at 20 revolutions per second, when it is connected to 500 V supply for a particular load. What will be the speed of the motor, when the load is reduced by 50%. Neglect contact drop and magnetic saturation.

Let the particular load be I_{a1}

$N_1 = 20 \times 60 = 1200$ r.p.m.

$E_{b1} = \frac{\phi Z N_1 P}{60A}$

$= \frac{50 \times 10^{-3} \times 492 \times 1200 \times 4}{60 \times 4} = 492$ V

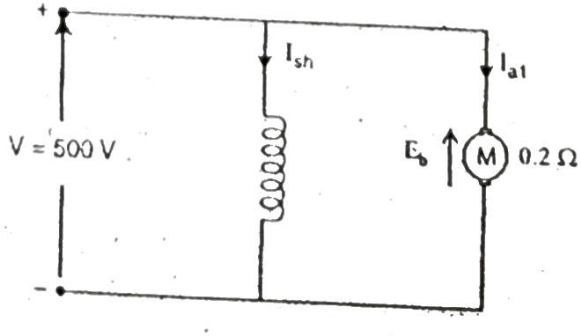


Fig.8.32

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$E_{b1} = V - I_{a1} R_a$ i.e. $492 = 500 - I_{a1} \times 0.2$, $\therefore I_{a1} = 40$ A

When $I_{a2} = 20$ A, $E_{b2} = 500 - 20 \times 0.2 = 496$ V

$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}}$ i.e. $\frac{1200}{N_2} = \frac{492}{496}$, $\therefore N_2 = 1209.76$ r.p.m.

8.24 The field current in a D.C. shunt machine is 2A and the line current is 20A at 200V. Calculate (i) the generated e.m.f. when working as generator (ii) torque in N–m when running at 1500 r.p.m. as motor. Take the armature resistance as 0.5 Ω.

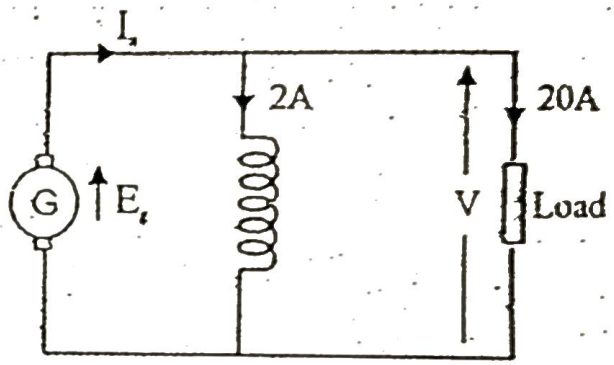


Fig.8.33

i) *As a generator*

$I_a = 20 + 2 = 22$ A

$E_g = V + I_a R_a$

$= 200 + 22 \times 0.5$

$= 211$ V

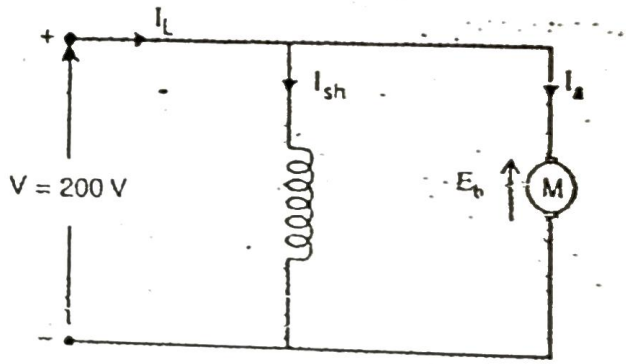


Fig.8.34

ii) *As a motor*

$I_a = I_L - I_{sh} = 20 - 2 = 18$ A

$E_b = V - I_a R_a = 200 - 18 \times 0.5 = 191$ V

$T_a = \frac{E_b I_a}{2\pi \frac{N}{60}} = \frac{191 \times 18}{2\pi \times \frac{1500}{60}}$

$= 21.9$ N–m

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NUMERICAL PROBLEMS

8.1 Determine the torque developed in the armature of a 500 V, 4 pole wave connected D.C. shunt motor having 860 conductors. The armature current is 100A, and the flux per pole is 25 mWb.

(683.7 N-m)

8.2 A 20 H.P., 4 pole, 250V, lap connected D.C. shunt motor has 48 slots, each containing 20 conductors. It draws a current of 80 A from the supply. The field and armature resistances are 125 Ω and 0.2 Ω respectively. The flux per pole is 100 mWb. Calculate (a) the speed (b) the torque developed by the armature (c) the shaft torque and (d) the rotational losses.

(146.5 r.p.m., 1,192.35 N-m, 959.32 N-m, 3,573.1 W)

8.3 A 500 V, 8 pole lap connected D.C. shunt motor running at 1500 r.p.m. has an output of 75 H.P. The armature and field windings have resistances of 1 Ω and 100 Ω respectively. The line current drawn is 200A. There are 100 slots, each containing 8 conductors. The drop at each brush is 1V. Find (a) useful torque (b) total torque (c) useful flux per pole (d) rotational losses and (e) efficiency.

(351.36 N-m, 376.147 N-m, 15.15 mWb, 3,835.05W, 55.163%)

8.4 A 300 V shunt motor with an armature resistance of 0.1 Ω is excited to give a constant main field. At full load, the motor runs at 1,000 r.p.m. and takes an armature current of 100 A. If a resistance of 0.5 Ω is placed in the armature circuit, find the speed at (a) full load torque (b) 1/2 full load torque and (c) Also find the ratio of starting torque to full load torque. **(827.59 r.p.m., 931.03 r.p.m., 5)**

8.5 A 230 V, D.C. series motor has a field winding and armature resistance of 0.2 Ω each and draws a full load current of 100 A and running at 1,000 r.p.m. Calculate the speed of the motor at which it develops twice the full load torque.

(645.45 r.p.m.)

8.6 An 8 pole, 600 V D.C. shunt motor takes 10A on no load at 1,500 r.p.m. The shunt field current is 4A. Calculate the full load speed of the motor, if it takes 150 A at full load. $R_a = 0.2\ \Omega$, Contact drop per brush is 1V. The armature reaction weakens the flux by 2% on full load.

(1,458.8 r.p.m.)

8.7 A 6 pole, D.C. shunt motor connected to bus bars of voltage 250 V takes a current of 5 A on no load and is running at 1000 r.p.m. Armature and shunt field resistances are 0.1 Ω and 250 Ω respectively. If the motor draws 51 A at a certain load, what will be its E.M.F. and speed?

(245V, 981.57 r.p.m.)

8.8 A 4 pole, 400 V, wave connected, D.C. shunt motor takes a full load current of 1,000 A at 1,000 r.p.m. There are 640 armature conductors and armature resistance

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is 0.2 Ω. The shunt field resistance is 200 Ω. Total brush contact drop = 2V. Find the useful flux per pole. **(17.83 mWb)**

8.9 A 300 V shunt motor takes a line current of 80 A and runs at 1,200 r.p.m. The field resistance and armature resistance are 100 Ω and 0.2 Ω respectively. Contact drop per brush = 1 V. Calculate (a) the no load speed, when the no load current is 10 A (b) the percentage reduction in the flux per pole, in order that the speed may be 1,500 r.p.m., when the armature current is 50A. Ignore the effects of armature reaction. **(1,259.28 r.p.m., 32.63%)**

8.10 A 250 V D.C. series motor has a field resistance of 0.1 Ω and an armature resistance of 0.2 Ω and runs at 1000 r.p.m. The current taken from the supply is 50 A. Find the speed at which it will run when connected in series with a 2 Ω resistance, and taking the same current. **(574.49 r.p.m.)**

8.11 An 8 pole, 440 V D.C. shunt motor takes 10 A on no load and runs at 1500 r.p.m. The shunt field resistance is 110 Ω. Calculate the full load speed of the motor, if it takes 120 A at full load. The armature resistance is 0.1 Ω. Contact drop per brush=1V. Armature reaction weakens the field by 5% on full load. **(1539.24 r.p.m.)**

8.12 A 4 pole, 300 V D.C. shunt motor has a lap connected armature with 200 single turn coils. The armature resistance is 0.2 Ω. The armature current is 15A at a no load speed of 1,000 r.p.m. Find the useful flux per pole. **(45.55 mWb)**

8.13 A 50 kW, belt driven D.C. shunt generator running at 1500 r.p.m. on 400 V bus bars continues to run as a motor when the belt breaks, then taking 5 kW. What will be its speed? $R_a = 0.02\ \Omega$, $R_{sh} = 100\ \Omega$. Total brush contact drop = 2V. Ignore armature reaction. **(1,474.97 r.p.m.)**

8.14 A 230 V, D.C. shunt motor draws an armature current of 30A, when running at 1200 r.p.m. against a full load torque. $R_a = 1\ \Omega$. What resistance must be inserted in series with the armature to reduce the speed to 1,000 r.p.m. at the same torque? What will be the speed, if the load torque is doubled with this resistance in circuit? Assume the flux to remain constant throughout. **(620.39 r.p.m.)**

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QUESTION BANK

1. Explain the working principle of a D.C. motor.
2. What is the significance of back E.M.F. in a D.C. motor?
3. Derive the condition at which the efficiency of a D.C. motor is maximum. Explain why D.C. motors are not normally designed for this condition.
4. Derive an equation for the torque developed in the armature of a D.C. motor.
5. What is useful torque in a D.C. motor? How is it related to the speed and H.P. of the motor?
6. Give the classification of D.C. motors giving the respective relations between the applied voltage and the various voltage drops.
7. Explain why a D.C. series motor should never be started without load.
8. Explain what happens when the field winding of D.C. series motor is opened, when it is in running condition.
9. Sketch the various characteristics of D.C. motors and mention their applications.
10. Explain why a starter is required for a D.C. Motor and with a neat sketch, give the construction and working of a three point starter.

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